

The two founding duplications, and what “orthogonal” rests on

Ancestral OR repertoires · human tree · density fill, no experimental overlay (ancestral cut $p \geq 0.8875$). Node labels follow the main figure; repository names in brackets. Source: notebooks/ancestral/05.ten_node_lineage.ipynb.

1 · What dup 1 and dup 2 are

A duplication is a **node**; the chemistry is measured on the **branches leading away from it**. Node 0 has two daughter branches, so the first duplication is two edges, not one, and they behave completely differently. This is the distinction the phrase “dup 1” can hide.

Duplication 1 — node 0 [node_1], the gnathostome OR ancestor

The single ancestral receptor duplicates into the two class ancestors. Both daughter branches are shown; only the Class I branch carries a chemical signal.

branch	ligands	retained	gained	lost	what happens
0 → 1 [node_1 → node_2] Class I ancestor, 62 leaves	21 → 21	4	17	17	near-total turnover: 17 of 21 ligands replaced. This is the acid branch.
0 → 2 [node_1 → node_3] Class II ancestor, 371 leaves	21 → 8	7	1	14	a contraction. One molecule gained — no chemistry is testable on this branch.

So the first duplication is **asymmetric**. The Class I daughter rebuilds its repertoire; the Class II daughter loses two thirds of it and adds almost nothing. When the text says “the first duplication took acids”, the precise statement is **the Class I daughter branch of the first duplication**, i.e. $0 \rightarrow 1$.

Duplication 2 — node 2 [node_3], the Class II ancestor

Node 2 splits into the clade containing every other Class II gene and a single receptor, OR2AT4 (9606.A6NND4). That singleton sits on a long branch beside the outgroup — long-branch-attraction geometry rather than biology — and its placement is unstable, so it is set aside throughout and the duplication is read as $2 \rightarrow 5$. Nothing depends on it: it retains 4 of node 2’s 8 ligands and **gains none**, so it contributes no chemistry to any test here.

branch	ligands	retained	gained	lost	what happens
2 → 5 [node_3 → node_6] Class II clade, 370 leaves	8 → 27	7	20	1	an expansion, almost pure gain. This is the aromatic branch.

The chemistry on the two branches

gain set	n	carboxylic acid	aromatic ring
0 → 1 gains (Class I)	17	10/17 = 59% · OR 13.0 · P = 2×10^{-6}	5/17 = 29% · OR 1.03 · P = 1.0 (n.s.)
2 → 5 gains (Class II)	20	0/20 = 0% · OR 0 · P = 0.15 (n.s.)	15/20 = 75% · OR 7.9 · P = 2×10^{-5}
background: full odorant panel	754	11.0%	28.9%

A clean double dissociation: each branch is enriched for one group and flat or absent for the other. Fisher exact, two-sided, background = the full 754-molecule panel. Against the alternative background (the 44-molecule ten-node universe) the dissociation is identical but the odds ratios move to 4.3 and 8.2, so **the background must be stated wherever an OR is quoted**.

One asymmetry worth keeping in the text

standing repertoire	n	% acid	% aromatic
node 0 [node_1]	21	0%	62%
node 1 [node_2] · Class I	21	48%	33%
node 2 [node_3] · Class II	8	0%	50%
node 5 [node_6] · Class II clade	27	0%	70%

Acids and aromatics are not symmetric acquisitions. **Acids are genuinely new** — absent from node 0 (0%) and from the whole Class II side, they appear only on the Class I branch (48%). **Aromatics are re-concentrated, not invented** — node 0 is already 62% aromatic, well above the 29% panel background; node 2 dips to 50% and node 5 climbs to 70%. The defensible claim therefore concerns the **gains** on each branch, which is what the enrichment test

and the orthogonality test both measure.

2 · What the orthogonality analysis is

The functional-group tests above ask about two groups chosen in advance. The orthogonality test asks a different and stronger question: **did the two branches move the repertoire in different directions in chemical space overall**, using no pre-chosen feature. “Orthogonal” is used here geometrically and quantitatively, not as a synonym for “different”.

Statistic and coordinates

Coordinates. The reference set is the full panel P of 754 odorants. Each molecule i is described by 13 RDKit descriptors — MolWt, LogP, TPSA, HBD, HBA, rotatable bonds, aromatic rings, FracCsp3, heavy atoms, ring count, MolMR, BertzCT and oxygen count — giving a raw matrix X of 754×13 . Every column is then z-scored **over the whole panel**:

$$z_{ij} = (x_{ij} - \mu_j) / \sigma_j \quad \text{with } \mu_j, \sigma_j \text{ the mean and population SD (ddof = 0) of descriptor } j \text{ across all 754 molecules.}$$

A molecule is therefore a point in $\mathbf{R}^{13}$ whose coordinates are in units of panel standard deviations. Standardising on the panel rather than on the gain sets is what puts the two branches on one common scale and stops a descriptor dominating through its unit (BertzCT ranges in the hundreds, FracCsp3 in [0,1]).

One vector per branch. For a branch with parent node p and gain set G , the vector is a **difference of centroids** — where the gained molecules sit, minus where the parent's standing repertoire R_p already sat:

$$\mathbf{v} = \text{mean}_{i \in G}(\mathbf{z}_i) - \text{mean}_{i \in R_p}(\mathbf{z}_i)$$

vector	gain set G	reference R_p	length $ \mathbf{v} $
$\mathbf{v}_1$ branch 0 $\rightarrow$ 1	17 molecules gained	node 0's 21 ligands	3.34
$\mathbf{v}_2$ branch 2 $\rightarrow$ 5	20 molecules gained	node 2's 8 ligands	3.73

The vector is left **unnormalised**: its direction is the quantity of interest, its length (in panel SD units) records how far the branch moved. Each branch is referenced to **its own** parent, so each vector answers “where did this branch move relative to where it started”, not relative to a shared origin.

The statistic. The statistic is the **cosine similarity** between the two vectors; the angle is its readable form, and being a monotone transform of the cosine it carries exactly the same information:

$$\cos \theta = (\mathbf{v}_1 \cdot \mathbf{v}_2) / (|\mathbf{v}_1| |\mathbf{v}_2|) = -0.116 \rightarrow \theta = \arccos(\cos \theta) = 96.7^\circ$$

For the significance test the quantity compared against the null is $|\theta - 90^\circ|$, i.e. how close to orthogonal the pair is. Observed $|96.7 - 90| = 6.7^\circ$, so a null draw counts as at least as extreme when its angle falls in $[83.3^\circ, 96.7^\circ]$. No draw did.

The null. 2,000 draws (seed 0). Each draw samples a random gain set of 17 molecules and one of 20 **without replacement from the same 754-odorant panel**, keeps the two real parent centroids unchanged, and recomputes θ by the formulas above. Only the identity of the gained molecules is randomised — set sizes, references, descriptors and standardisation are all held fixed.

The two-dimensional panel is exact, not a projection. The figure draws both vectors in the orthonormal basis returned by the QR decomposition of $[\mathbf{v}_1, \mathbf{v}_2]$. That basis spans precisely the plane containing the two vectors, so the angle on the page equals the angle in $\mathbf{R}^{13}$ to the digit (96.67° either way). Nothing is lost in the drawing.

Should losses count?

The vector above is built from what a branch **gained**. That is a real omission, because a branch also loses ligands, and on the Class I branch the loss is not a footnote: it drops **17** of node 0's 21 ligands, as many as it gains. Three definitions were therefore compared, the second and third of which account for loss.

definition of the shift	angle	null median	draws $\geq$ obs
A gained – parent repertoire (as reported)	96.7°	33.3°	0 / 2,000
B child repertoire – parent repertoire (loss-aware)	98.4°	35.8°	0 / 2,000

C gained – lost (turnover contrast)	106.6°	89.5°	566 / 2,000 n.s.
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The conclusion survives accounting for losses. Definition B is the natural loss-aware statistic — the child repertoire is what the parent had, minus what went, plus what came — and it gives the same answer as A, against the same kind of null. **Definition C should not be used.** Differencing two independently drawn sets produces a noise-dominated vector, and two such vectors are near-orthogonal by construction: its null median is already 89.5°, so the observed 106.6° carries no information. That is a property of the statistic, not a result about the tree.

A is kept as the primary definition because the claim being made is about what each duplication *took up*; B belongs beside it as the check that this is not an artefact of ignoring what was dropped.

What losses add on their own

Looking at the discarded ligands, which no earlier version of this analysis did, sharpens the asymmetry between the two branches into a difference of **mode**:

	retained	gained	lost	mode
Class I 0 → 1	4 of 21 (19%)	17	17	replacement — the repertoire is rebuilt
Class II 2 → 5	7 of 8 (88%)	20	1	accretion — the repertoire is extended

One sentence to avoid. It is tempting to write that the Class I branch traded the ancestor's aromatics for acids — the lost set is 65% aromatic and the gained set 59% acid. The trade reading does not hold: **the loss was not chemically selective**. Within node 0's own repertoire, 11 of 13 aromatics were dropped against 6 of 8 non-aromatics (OR 1.83, $P = 0.62$) — the branch discarded most of what it had, indiscriminately, retaining only 19%. The aromatics went because nearly everything went.

The defensible statement is therefore two-part and asymmetric in kind: **a wholesale, chemically non-selective loss, followed by a selective gain of acids** (OR 13.0, $P = 2 \times 10^{-6}$). Only the gain carries chemical information, which is also why the gain-based vector is the right primary definition.

Robustness of the geometry

The 13 descriptors are correlated (max $|r| = 0.98$ for MolWt / heavy atoms / MolMR; mean $|r| = 0.38$), and a Euclidean angle in a correlated space effectively upweights whichever block is most redundant. The check is to repeat the whole computation in a decorrelated space:

variant	angle	null median	draws $\geq$ observed
as specified (Euclidean, parent-referenced)	96.7°	33.3°	0 / 2,000
whitened (Mahalanobis, panel correlation matrix)	88.5°	45.6°	0 / 2,000

Whitening moves the angle *towards* 90°, not away from it, and no null draw reaches it under either metric — so the result is not an artefact of descriptor redundancy.

The choice of reference does matter and should be stated when the number is quoted. Measuring both branches against node 0 gives 108°; using the raw gain centroids with no parent subtraction gives 73°. Both remain far above the null median of 33°, but they are different questions; the pre-specified definition — each branch against its own parent — is the one reported throughout.

Result

quantity	value	reading
observed angle	96.7°	cosine similarity -0.116 — essentially independent directions
null median	33.3°	IQR 27.8–39.9°; maximum over 2,000 draws = 78.2°
null draws as close to 90°	0 / 2,000	$P < 0.0005$; the observed angle exceeds every draw
leave-one-descriptor-out	90.2–102.5°	no single descriptor produces the result
$\phi(\text{acid, aromatic})$ in the panel	-0.10	16% of acids are aromatic, 6% of aromatics are acids

Why the null is the point

The null is **not** 90°, and this is what makes the result non-trivial. Most odorants in this panel share bulk chemistry, so any two shifts tend to point broadly the same way. Drawing random gain-sets of the same sizes (17 and 20) from the panel, keeping the same two parent centroids, and recomputing the angle gives a median of **33°**. The observed 97° is not merely above that median — it is above the largest of 2,000 draws (78°). The two branches moved the repertoire in directions that random gains of the same size essentially never produce.

The double dissociation is also not definitional. Acid and aromatic are near-independent features in this panel ($\phi = -0.10$), so gaining acids does not mechanically preclude gaining aromatics — a branch could have taken both.

As an internal check, the vectors' own composition matches the functional-group result without having been told to: **v**₁ loads on HBD (+1.80), TPSA (+1.58) and rotatable bonds (+1.30) — the carboxylic-acid signature; **v**₂ loads on aromatic rings (+1.07) and HBA (+0.87) against LogP (−1.59) and FracCsp3 (−1.51) — the aromatic signature.

Limits to carry into the text

Quote the verdict, not the digits. The exact angle depends on which descriptors are in the set: leave-one-out spans 90–102°, and an earlier version of this analysis on a different 13-descriptor set reported 86.6° with a null median of 43°. Write “close to orthogonal (~90°), against a null median of ~33° and beyond all 2,000 random draws” rather than a single figure to one decimal.

n = 17 and n = 20. These are small sets. The acid and aromatic enrichments survive because the effects are very large, and the angle survives because its null is explicit — but no finer chemical or perceptual claim about these two branches is powered at this sample size, and none should be made.

Reconstruction, not observation. Every ligand call on an ancestral node is a model prediction under the density fill; no measurements exist for a reconstructed sequence. The result is a statement about the reconstructed repertoires, conditional on the tree and the binarisation.

3 · What “orthogonal” means here, in plain terms

The word invites one specific misreading, so it is worth separating two things that sit next to each other in the analysis and are **not** the same claim.

	what it is about	what it says	role
$\phi = -0.10$ acid vs aromatic	the molecules in the panel	being an acid and being aromatic are near-independent properties of an odorant	a precondition , not the result
$\theta = 97^\circ$ the angle	the two branches	the direction each branch moved the repertoire's chemistry is independent of the other's	the result

So no — orthogonality is not the statement that acids and aromatics are uncorrelated. That fact is about molecules and it is used only to rule out a trivial explanation: if every acid were aromatic, a branch taking acids would be forced to take aromatics too, and the dissociation would be an artefact of chemistry rather than a fact about evolution. $\phi \approx 0$ says that trap is not sprung. The angle is a separate measurement about the two evolutionary changes.

The plain reading

Picture each branch as an **arrow** describing how it shifted the average chemistry of the repertoire: which way, and how far. The question is what the two arrows do relative to each other. Pointing the same way (0°) would mean both duplications pulled the repertoire in the same chemical direction. Pointing oppositely (180°) would mean one undid the other. **At right angles means neither: knowing what the first duplication did to the repertoire tells you nothing about what the second did.** The two events explored chemical space independently.

The reason this is a finding rather than a truism is the null. One might expect two unrelated changes to be at right angles by default — but they are not. Because most odorants in this panel share bulk chemistry, two *random* sets of gains pull the repertoire in broadly the same direction: the median angle between random shifts is **33°**, and none of 2,000 draws reached 90°. Against that background, two branches at right angles is the unusual outcome, not the expected one.

Is it just the acids-and-aromatics result restated?

This is the obvious objection, and it was tested rather than assumed. If the angle were simply the acid/aromatic dissociation expressed in geometry, it should disappear once those two features are taken out of the descriptor space. It does not:

descriptor space	angle	null median	draws $\geq$ observed
all 13 descriptors (as reported)	96.7°	33.3°	0 / 2,000
the 5 acid- and aromatic-carrying axes removed	114.0°	—	—
acid and aromatic membership regressed out of every descriptor	107.8°	30.4°	0 / 2,000

The separation therefore extends **beyond the two functional groups the enrichment tests name**. After removing everything acid-ness and aromaticity contribute, the two branches still moved in directions at or past right angles, and still no random draw comes close. This is the strongest argument for keeping the analysis: it shows the two duplications took different chemistry in a broader sense than “one took acids, the other aromatics”.

Simpler ways to show the same thing

approach	result here	how it reads
PCA of the panel with the two gain sets and their shift arrows drawn on it	PC1 42%, PC2 21%; the two arrows are visibly at right angles	the most straightforward, and the recommended display. A reader sees two arrows leaving their parent repertoires in different directions, with no geometry to decode. See the caveat below.
Double dissociation (two Fisher tests)	acids 59% vs 0%; aromatics 29% vs 75%	the best powered and the plainest sentence. Limited to the two features chosen in advance, which is exactly what the geometry removes.
PERMANOVA on the two gain sets (Euclidean, 4,999 permutations)	pseudo-F = 3.20, $R^2 = 0.084$, P = 0.011	the multivariate test this readership expects, so it needs no explaining — but markedly weaker here, and it answers “are the two sets different?”, not “were the two changes independent?”

The caveat the PCA panel must carry. A PCA plane is a projection, so unlike the exact two-vector panel the angle a reader measures off the page is *not* the statistic. Here the first two components hold 64% of the variance and capture **67%** of the Class I shift and **88%** of the Class II shift; the projected angle reads 106° against a true 96.7°. The picture is therefore faithful in direction but flattering in magnitude, and the caption should give the 13-dimensional number rather than inviting the reader to measure the drawn one. A dispersion check confirms the PERMANOVA reflects a location difference rather than one group being more scattered (mean distance to own centroid 6.6 vs 2.9, Levene P = 0.38).

Recommended framing for the manuscript

Lead with the double dissociation in the main text and figure — it is the clearest statement of the result and the best powered. **Put the angle in the supplement** with its null and the control above. It is a non-standard statistic; used as the headline it makes a reader stop and decode geometry, whereas used as support it does real work by showing the effect is not confined to two hand-picked functional groups.

Draft wording that avoids the jargon entirely:

“The two founding duplications took up different chemistry: the Class I branch gained carboxylic acids (10 of 17 gained odorants; OR 13.0, $P = 2 \times 10^{-6}$) while the Class II branch gained aromatics (15 of 20; OR 7.9, $P = 2 \times 10^{-5}$), with neither branch enriched for the other’s chemistry. Treating each duplication as a shift in a 13-descriptor chemical space, the two shifts were close to independent in direction (angle 97°, versus a median of 33° for random gains of the same size; 0 of 2,000 permutations as close to 90°), and remained so after acid and aromatic character were regressed out (108°) — indicating the two lineages sampled distinct regions of odorant chemical space rather than dividing a single axis of variation.”

4 · What the observation means biologically

Three separate things happen at these nodes, and they are easy to run together. Kept apart, they make one coherent account of how a single ancestral receptor became a two-class, combinatorially-read repertoire.

a. The combinatorial code is a by-product of duplication, not an adaptation

One receptor cannot produce a combinatorial code: every odorant it detects is, trivially, detected by one receptor. Two can, and immediately do — 7.4% of what the pair detects is read by both, rising to 32% at three. No selective explanation is required, and none should be offered: duplicating a broadly-tuned receptor produces overlap for free.

The non-trivial part is the *size* of the early overlap. Receptors independent at their observed breadths would give roughly 0.3% and 1% at two and three receptors, against the observed 7.4% and 32% — so the early code is far **more** overlapping than chance. That excess is inherited breadth: the ancestor was a generalist, and its daughters started as generalists too.

b. Overlap and divergence happen at the same time, on different molecules

The apparent tension — receptors that overlap more than chance, yet whose gains point in independent chemical directions — dissolves once the repertoire is split into what was **kept** and what was **added**.

	what it is	what it produces
the retained core	node 0's ligands, held by both daughters and never lost; at the present they are the most redundantly detected odorants in the repertoire (median 95 detectors against 7 for everything else)	the combinatorial code — many receptors reading the same molecules
the divergent additions	what each branch gained on top: acids on the Class I branch, aromatics on the Class II branch, in independent directions and independently of those two groups	coverage and discrimination — new chemistry each lineage alone can detect

So duplication did not simply make two copies that drifted apart. It produced a **conserved generalist core plus lineage-specific extensions**, and the two outcomes that look contradictory — a code that is combinatorial and classes that are chemically distinct — are the two halves of that single architecture. This is the sentence the analysis actually earns.

c. The two classes did not divide one axis of chemistry between them

The naive reading of a duplication splitting a repertoire is partitioning: the ancestor's chemistry is cut in two, one daughter taking each half. That is not what happened, and the orthogonality is what rules it out. Partitioning a single axis would leave the two shifts pointing in *opposite* directions along that axis (near 180°). Right angles mean the two lineages left the ancestral condition along **different axes altogether**, each exploring chemistry the other did not.

The two branches are also strikingly asymmetric in form:

	Class I branch (0 → 1)	Class II branch (0 → 2 → 5)
what the branch does	rebuilds: 17 of 21 ligands replaced in one step	contracts to 8 ligands, then the surviving clade expands to 27
chemistry taken	carboxylic acids — absent from the ancestor (0%) and from the entire Class II side, so a genuine acquisition	aromatics — the ancestor is already 62% aromatic, so a re-concentration (50% → 70%), not an invention
subsequent fate	stays small: saturates near 62 receptors	radiates: 371 of the 433 human receptors

Read together: **the Class I lineage innovated and stopped; the Class II lineage bottlenecked, then elaborated chemistry it already had**. The class division is therefore not a symmetric split of an ancestral niche but two different evolutionary modes — acquisition of a new chemical capability on one side, and amplification of an existing one on the other.

The acid association is worth noting because it is not something the reconstruction was told. Class I receptors are the clade humans share with fish — hence “fish-like” — and are independently associated with carboxylic acids (the OR51 and OR52 families and short-chain fatty acids). That the reconstruction places an acid acquisition on the founding Class I branch is an external consistency check on the method rather than a new claim, and the primary literature should carry it.

What this does not license

No timing. The order — acids at the first duplication, aromatics at the second — is a statement about tree topology, not dates. The two time axes bracket the timing rather than fixing it, and a human-only tree cannot narrow it.

No selection. Nothing here distinguishes chemistry that was selected for from chemistry that drifted and was retained. The orthogonality says the two lineages ended up exploring independent directions; it does not say anything pushed them there.

No perception. These are predicted receptor–odorant interactions. At $n = 17$ and $n = 20$ no odour-descriptor or perceptual claim is powered, and the sets are too small for any statement about what these ancestors could *smell* as opposed to bind.